Retail Growth and Profitability Comparison

Amazon North America and Walmart U.S. | 2019-2024

Project purpose

This project compares sales growth and operating profitability for an internet-origin retailer and a store-origin retailer. The 2019 pre-pandemic reference helps describe changes during 2020-2021 and the subsequent 2022-2024 adjustment period.

Data and scope

The sample contains twelve annual segment records: six per business. Segment net sales and operating income were manually transcribed from free official annual reports into financial_data.csv. Amounts are nominal USD millions. Actual reporting dates and source references accompany each record. Amazon North America excludes AWS; Walmart U.S. excludes Sam's Club and Walmart International.

Four-module structure

Module	Main task	Class	Output
1. Research Configuration	Load research settings, fiscal offsets, labels and sources from config.json.	StudyConfig	Shared settings
2. Data and Database Management	Read the CSV, clean amounts, align years and preserve dates and sources.	DatabaseManager	Financial records in SQLite and CSV
3. Metrics and Comparative Analysis	Calculate annual measures, sales CAGR and stage means; query stage growth in SQL.	MetricCalculator	Metrics, CAGR and stage tables
4. Charts and Reporting	Read the results and combine charts, findings, context and source links.	ReportBuilder	Three charts and an HTML report

Construction logic

Official filings > financial_data.csv > raw_df > clean_df > SQLite financial_records > financial_df > annual_metrics and summaries > charts and report. The configuration file supplies the research choices at each step. Small classes store shared settings or output locations; ordinary functions perform cleaning and stage summaries.

Content and framework separation

The notebook contains reusable processing and presentation rules. Company details, chart choices, source references and business commentary live in config.json; financial observations live in the input CSV. A different comparison requires changing these inputs, including the actual data and commentary. The same framework applies when the input schema and financial measures remain suitable.

Measures and core results

Measure	Calculation
Annual sales growth (%)	(Current sales / previous annual sales - 1) x 100
Operating margin (%)	Segment operating income / segment net sales x 100
Sales index	Sales / 2019 segment sales x 100
Sales CAGR, 2019-2024 (%)	[(2024 sales / 2019 sales) raised to (1/5) - 1] x 100

Six annual observations contain five compounding intervals. CAGR measures annualized compound sales growth; it differs from cumulative sales change and the arithmetic mean of annual growth. The 2019 growth value is missing because 2018 is not included. Negative operating income is retained.

Comparison with the 2019 reference

Segment	Sales change 2019-2024 (%)	Sales CAGR 2019-2024 (%)	Margin (%) 2019 to 2024	Margin change (points)
Amazon North America	+126.91	17.81	4.12 to 6.44	+2.32
Walmart U.S.	+35.60	6.28	5.10 to 5.16	+0.07

Amazon North America grew faster over the full window, but its margin fell to -0.90% in 2022 before reaching 6.44% in 2024. Walmart U.S. grew more slowly, with margins remaining close to 5%. Amazon's 2024 margin exceeded Walmart's 5.16%.

Fiscal-year alignment

Analysis year 2023 pairs Amazon CY2023 (January 1-December 31, 2023) with Walmart FY2024 (February 1, 2023-January 31, 2024). Thus CY2019-CY2024 is paired with FY2020-FY2025. Each window is annual, but Walmart's is shifted one month later; this is approximate alignment, not identical coverage.

Stage comparison and limits

Stages are 2019 baseline, 2020-2021 disruption and 2022-2024 adjustment. Stage means average available annual values and include valid-year counts. SQL AVG and COUNT reproduce the stage growth summary directly from SQLite, excluding NULL growth. The comparison is descriptive: one pre-pandemic year, different geography, business mix and revenue recognition, and no inflation adjustment limit causal interpretation.

Financial data sources

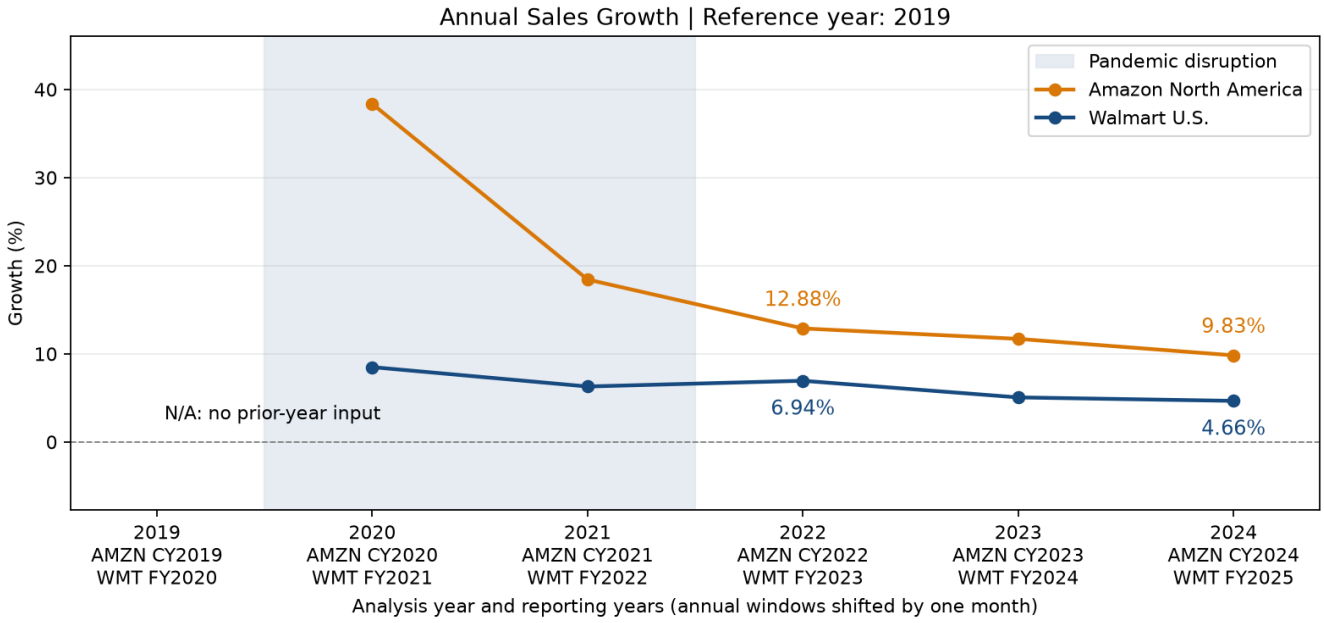
Amazon 2021 Form 10-K, Note 10: North America, 2019-2021.

Amazon 2024 Form 10-K, Note 10: North America, 2022-2024.

Walmart FY2022 Form 10-K, Item 7: Walmart U.S., FY2020-FY2022.

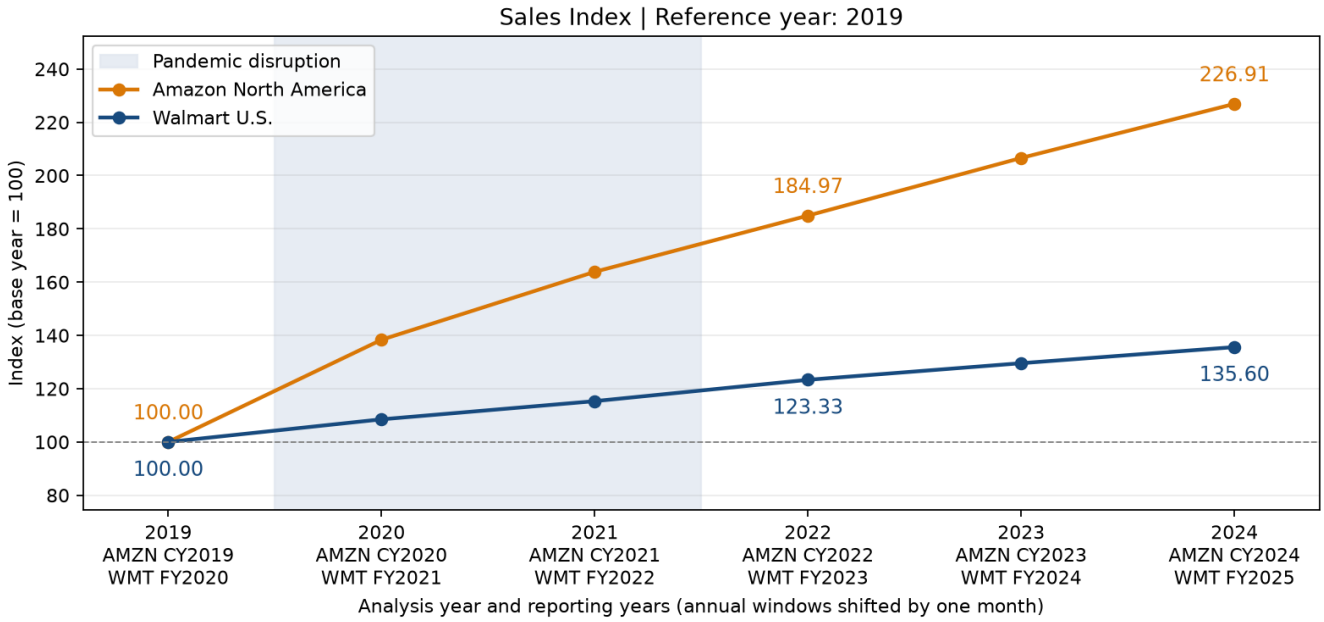
Walmart FY2025 Form 10-K, Item 7: Walmart U.S., FY2023-FY2025.

Sales growth and sales scale



Source: official segment filings. CY = calendar year; FY = fiscal year.

Figure 1. Annual sales growth. The 2019 value is N/A, not zero. Key-year labels show 2022 and 2024 values; the shaded area marks 2020-2021.

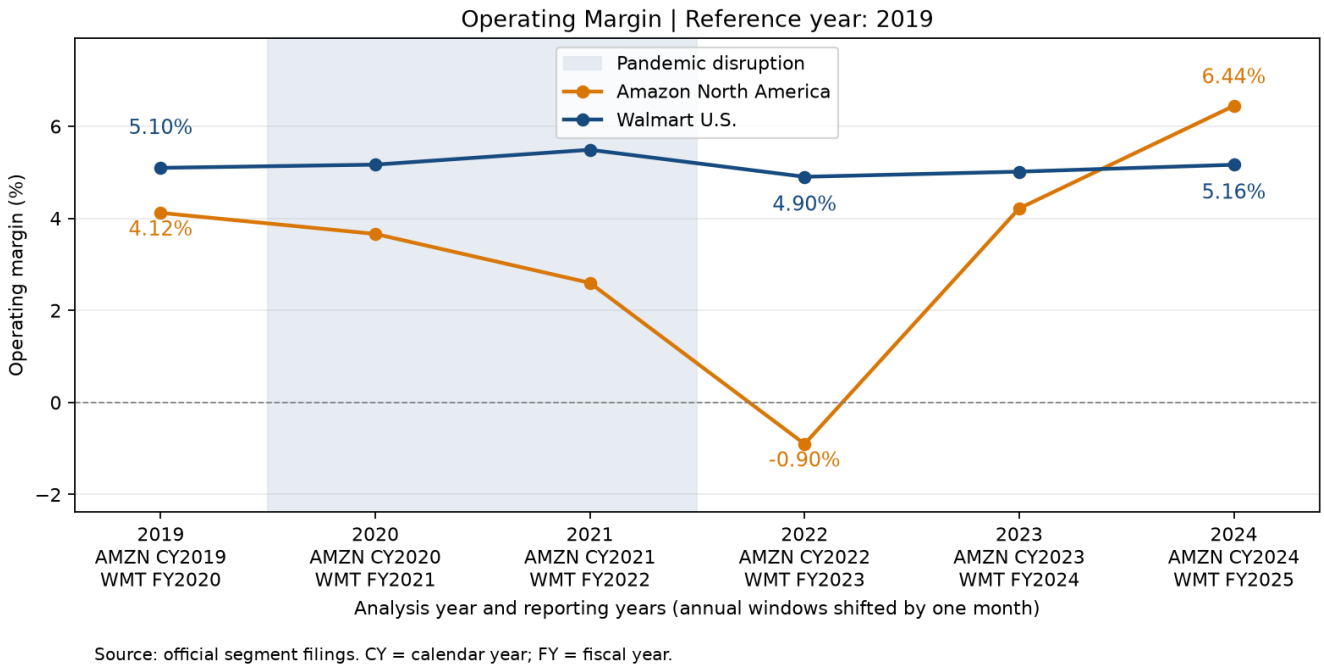


Source: official segment filings. CY = calendar year; FY = fiscal year.

Figure 2. Each segment's 2019 sales equal 100. The 2024 indexes are 226.91 for Amazon and 135.60 for Walmart. The corresponding five-year sales CAGRs are 17.81% and 6.28%.

Each tick pairs Amazon calendar years with Walmart fiscal years. The shaded period is a study grouping and does not identify a causal pandemic effect.

Operating profitability



Source: official segment filings. CY = calendar year; FY = fiscal year.

Figure 3. Segment operating income divided by segment net sales. The 2019, 2022 and 2024 values are labeled; negative margins represent an operating loss.

The 2022 margin dip

Amazon rapidly expanded its fulfillment network during the pandemic. Its 2022 filing describes demand returning toward normal seasonal levels relative to network fixed costs, alongside higher transportation, utility and wage costs. This supports an interpretation of capacity growing faster than normalized demand and inflation-related cost pressure around the North America loss.

Amazon 2022 Form 10-K, MD&A: operating cost pressures

Amazon 2022 shareholder letter: fulfillment expansion

The recovery through 2024

Amazon describes regionalization as placing inventory closer to customers, shortening transport distances and lowering cost to serve; its 2024 letter describes continued network improvements. These changes provide context for the margin recovery. The 2024 filing also attributes higher North America operating income primarily to increased unit and advertising sales, partly offset by higher total fulfillment and shipping costs. Lower unit costs do not mean total costs fell.

Amazon 2023 shareholder letter: regionalization and unit costs

Amazon 2024 shareholder letter: fulfillment network redesign

Amazon 2024 Form 10-K, MD&A: North America operating income

These management explanations provide business context. This small descriptive sample does not isolate the causal contribution of each factor. Lower unit costs and higher total costs can coexist as sales volumes grow.